A Cork Ball

Green Apple

If you throw each of the spherical shaped object, do you think each time the spherical shape is going to behave in the same way? Why? With the same force and weight, if you throw each one which one will go furthest?
Try to understand how they operate, what changes and how each one relates to the others?

A Cricket Ball

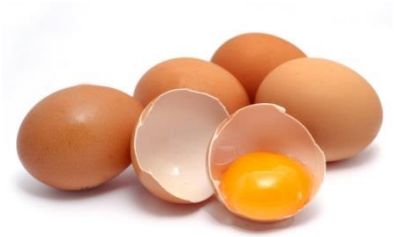
A Golf Ball

Plum

- Paradigm in Nature gives us handle on how to think of and talk about the ways that-
- **Things operate**
- **How they change, and**
- **How objects relate to others (metaphors)**
- Paradigm tries to understand some of the basic principles and their nature of functioning.

Manu Prakash at Stanford.mp4

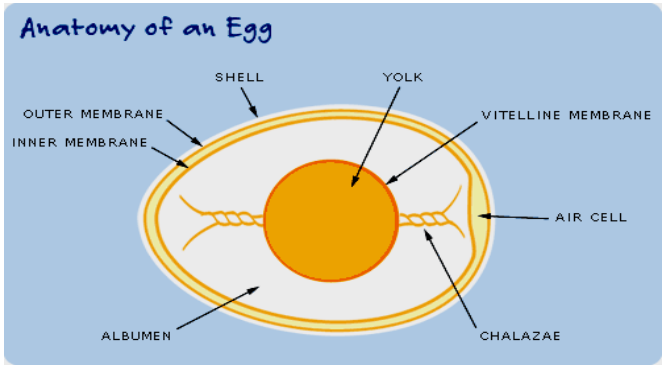
- **The course focuses on 'Innovation and Creativity'. Can you think of innovation without thinking about the fundamental scientific principles? The magic of scientific principles, if blended with the imagination (creativity) it can produce miracle.**
- **Dr. Manu Prakash a faculty at Stanford University Department of Biotechnology is producing such miracle that would affect the life of millions throughout the world. His thinking starts with simple basics not with high technology. He has curiosity like a child! He is excited about his work and doing some amazing innovative work at Stanford!**
- **Let us see what he does with so little yet so much…**
- **Couple of the links-** www.foldscope.com **/** http://microcosmos.foldscope.com



Syria-Beehive Structure

The structure keeps inside cooler

- **Simple Shape Paradigm**
- The simple-shape of paradigms are easily recognized, grasped, described, and appreciated for their simplicity and elegance.
- Simple-shape paradigms are often, made of single material.
- From its outer form may appear simple-shape paradigm, it might have complexity hidden within.
- The structure and the content of a chicken egg reveal the complexity of the shock absorbing design paradigm hidden in a simple (apparent) design. Similarly, the construction of the beehive shows the complexity of geodesic structure. The complexity of a centipede coiling into a disc like structure shows the nature's hidden strength through design paradigms.
- The simple-shape paradigms are important to understand the complex design. Simple-shape design shows visual and structural elements of forms that would constitute the complex form design.



The simple egg-structure has interlocking complex shapes

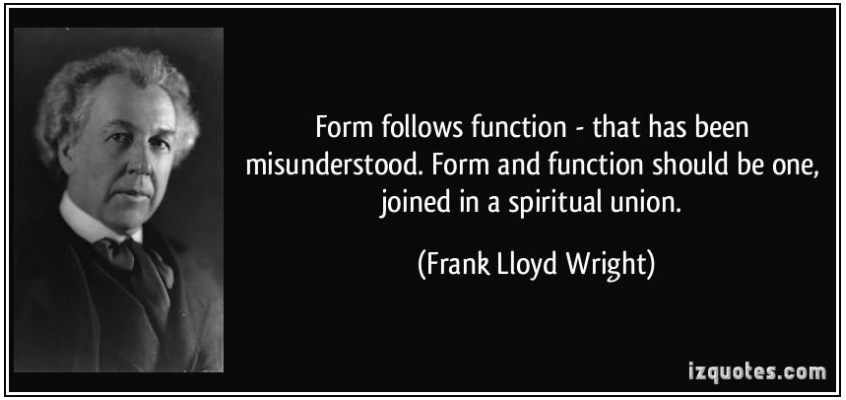
Leonardo's Sketch

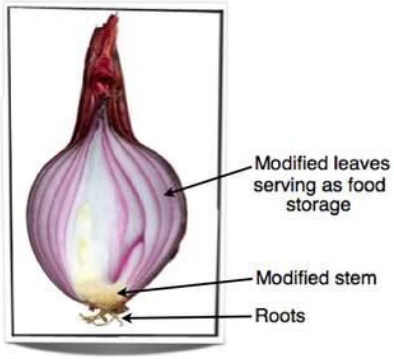
Complex shape

- **Simple Shape Paradigm**
- The simple-shape of paradigms are easily recognized, grasped, described, and appreciated for their simplicity and elegance. Where as the groups of paradigm involve interlocking, moving, or changing parts, or relationships between many parts.

- **Simple-shape paradigms are often, but necessarily, made of single material**.
- **Complexity of honeycomb structure in its simple –shape structure reveals the source of 'geodesic' dome structure (Buckminster Fuller)**

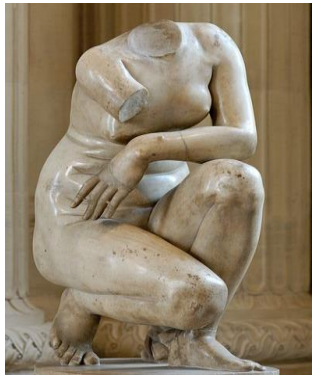
- The simple-shape paradigms are important to understand the complex design. **Simple-shape design shows visual and structural elements of forms that would constitute the complex form design.**

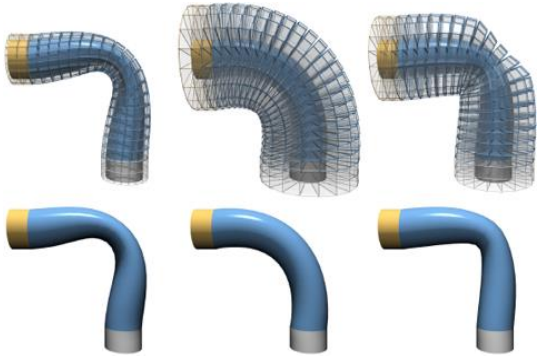




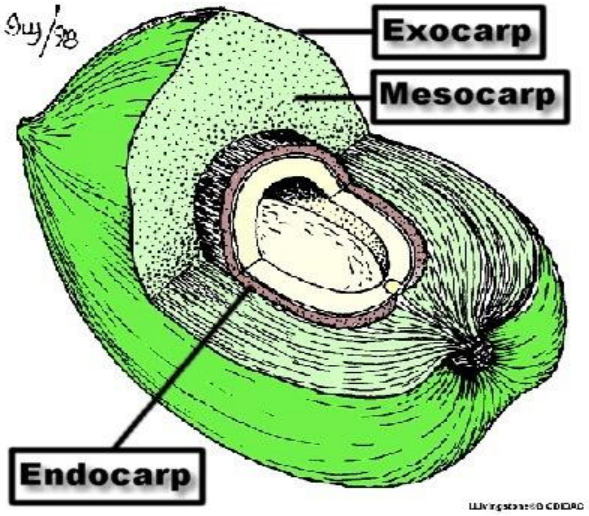
- **Where Does Form Come From?**
- A phrase popularized by 19th C architect Louis Sullivan (1856-1924) "**form follows function"** … form that expresses its environment, its structural system, and its purpose. Form has to reflect the environment and harmonize with the surroundings
- However, often architects have worked the other way- function follows the form.
- 'Form is independent from function'

- **Design has to understand several categories-**
- ***Basic forms*,***
- ***Functional relationships* (such as the way two parts relate to each other)**
- ***Behaviours* (such as the ways in which an object can get bigger or smaller).**





- **While deciding about the *form*, paradigm reveals the fundamentals of form. In nature each form is integral part of its function. Function is one of the most important factors in *form*.**
- **Basic Form- a collection of distinct useful forms;**
- **Functional Relationship- mechanism & technique among various parts relate to each other**
- **Behaviour- relationship between the changing forms (bigger& smaller, etc.)**



Coconut

Centipede Crawlling

Ladybug

Caterpillar Crawling.mp4

How Caterpillars Crawling .mp4

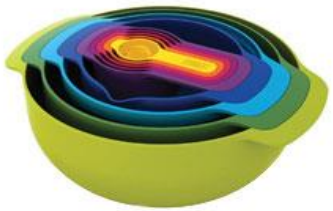
Tank Crawling up hill.mp4

Natural process is based on a complex harmonious relationship. The relationship between various living and non-living beings build the delicate but stable structure is the resource of the design process.

- # Design and Nature
- **Nature is the wealth of design solutions. Many of them are paradigmatic, such as layers of onion (nesting) or the way Kangaroo caries its baby in the pouch. Stacking spoons or dinner plates uses the paradigm of the construction of onion layers. Similarly, when we look at the way the planet orbits around the sun due to the gravitational force, principle of orbiting may be considered a paradigm. The same principle may be applied in a design process.**

- **Onion Shell Construction (nesting) Spoon stacked together (nesting)**
- ***Geodesic* structural**

NESTING



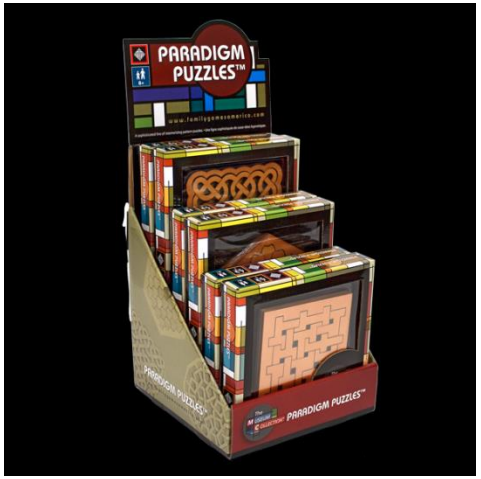
Nesting- manmade

Toy

Environmental Design

Art- Sculpture

Architecture



Book-rack

Furniture

Nesting

Pouch Carry Bag

Nesting in Nature

Stacking